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Chapter 10 — Human Respiratory System

Complete Chapter Notes · Textbook topics converted into board-style long questions, with important conceptual short questions inside every topic · PECTAA 2025–26, Punjab English Medium

This chapter follows the textbook topics 10.1 to 10.4. Each topic is converted into board-style long questions, and each topic ends with conceptual short questions. Direct definitions are omitted.

Board pattern (Punjab 2026, new syllabus): Chapter 10 carries 1 MCQ, 3 short questions in Q.4, and part (b) of a long question in Q.7. Shorts are 2 marks; longs are 8 marks.

TOPIC 10.1 — RESPIRATORY SYSTEM OF MAN

Textbook topic 10.1 · printed pages 215–221

Q.1 Describe the properties of the respiratory surface and the components of the human respiratory system. Explain the mechanism of breathing.

Board long question · 8 marks

Properties of respiratory surface

- It is **moist and permeable** so gases can pass through it.
- It is **thin** so gases travel a minimum distance.
- It has a **blood supply** so gases can diffuse in and out of blood.
- It has **structural support** so it remains open and does not collapse.
- It is **located internally** so its moist surface does not lose water to the atmosphere.
- Air **ventilates** over it, moving towards and away from it.
- Air reaches it after passing a **branched tubular way**, becoming saturated with water vapour.

Components

The respiratory tract has two divisions. The **upper respiratory tract** includes nasal cavity, pharynx and larynx; it cleans, humidifies and warms air but does not exchange gases. The **lower respiratory tract** includes trachea, bronchi, bronchioles and lungs, where gas exchange occurs.

The trachea has C-shaped cartilage rings to keep it open. Bronchi branch into bronchioles and end in **alveoli**, the functional units of gas exchange. Alveoli are moist, thin (about 0.1 µm) and surrounded by capillaries. The lungs are covered by visceral and parietal pleura, with pleural fluid allowing free expansion.

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Mechanism of breathing

- **Inspiration:** Diaphragm contracts and flattens; intercostal muscles raise ribs. Thoracic volume increases, lung pressure falls below atmospheric pressure, and air enters.
- **Expiration:** Diaphragm and intercostal muscles relax. Thoracic volume decreases, lung pressure rises above atmospheric pressure, and air leaves.

Important Conceptual Short Questions — Topic 10.1

S1. Why must the respiratory surface be thin and moist?

A thin surface minimizes diffusion distance, and moisture allows gases to dissolve before diffusing across the membrane.

S2. How do C-shaped cartilage rings help the trachea?

They provide strength and support to keep the airway open while allowing flexibility.

S3. Why are alveoli considered the functional units of the lungs?

They provide a large, thin, moist surface surrounded by capillaries where oxygen and carbon dioxide are exchanged by diffusion.

S4. How does surfactant help alveoli?

It reduces surface tension, preventing alveoli from collapsing and sticking together when air moves out.

S5. Why is the left lung smaller than the right lung?

It has two lobes instead of three, allowing room for the heart, which lies slightly left of the centre of the chest.

S6. How does contraction of the diaphragm cause inspiration?

It flattens and lowers, increasing thoracic volume and lowering lung pressure below atmospheric pressure, so air flows in.

S7. Why is breathing harder at high altitudes?

Atmospheric pressure is lower, so a greater increase in thoracic volume is needed to lower lung pressure enough for air to enter.

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TOPIC 10.2 — TRANSPORT OF GASES

Textbook topic 10.2 · printed pages 221–224

Q.2 Explain the transport of oxygen and carbon dioxide in the blood. What is the Bohr effect?

Board long question · 8 marks

Transport of oxygen

Oxygen diffuses from alveoli into pulmonary capillaries. A small amount dissolves in plasma, but most binds to **haemoglobin** in RBCs, forming **oxyhaemoglobin**. At alveolar pO_2 of about 105 mm Hg, about 97% of haemoglobin is saturated. In tissues, pO_2 is about 40 mm Hg, so oxyhaemoglobin releases oxygen and becomes deoxyhaemoglobin.

During exercise, venous pO_2 may fall to 20 mm Hg and haemoglobin saturation drops to 35%, so more oxygen is unloaded. The large oxygen reserve helps meet increased demand.

Bohr effect

CO_2 produced by tissues lowers blood pH. Lower pH reduces haemoglobin's affinity for oxygen, causing it to release oxygen more readily. Increased temperature has a similar effect, helping unload oxygen during exercise.

Transport of carbon dioxide

- **Bicarbonate ions (~72%):** CO_2 combines with water in RBCs to form carbonic acid, catalysed by carbonic anhydrase. It dissociates into H^+ and HCO_3^-. Bicarbonate moves into plasma in exchange for chloride ions (chloride shift).
- **Carbaminohaemoglobin (~20%):** CO_2 binds to the globin chains of haemoglobin. (The textbook writes "carboxyhaemoglobin"; the correct term for CO_2-bound haemoglobin is carbaminohaemoglobin.)
- **Dissolved in plasma (~8%):** A small amount of CO_2 is carried dissolved in plasma.

In the lungs, low pCO_2 reverses these reactions, and CO_2 diffuses into alveoli to be exhaled.

Important Conceptual Short Questions — Topic 10.2

S1. Why is most oxygen transported bound to haemoglobin rather than dissolved in plasma?

Plasma can dissolve only about 3 mL O_2 per litre, while whole blood carries almost 200 mL O_2 per litre because haemoglobin greatly increases oxygen-carrying capacity.

S2. How does the Bohr effect help deliver oxygen to active tissues?

CO_2 lowers pH, reducing haemoglobin's affinity for oxygen and promoting oxygen release where it is needed most.

S3. Why does exercise increase oxygen unloading from haemoglobin?

Active muscles use more oxygen, lowering venous pO_2, and produce CO_2 and heat, which reduce haemoglobin's affinity for oxygen.

S4. How does the chloride shift help carbon dioxide transport?

It allows bicarbonate ions to move out of RBCs into plasma in exchange for chloride ions, maintaining electrical balance while CO_2 is carried as bicarbonate.

S5. Why does carbon monoxide poisoning cause tissue hypoxia?

CO binds haemoglobin with much higher affinity than oxygen, reducing the amount of haemoglobin available to transport oxygen.

S6. Differentiate between oxyhaemoglobin and deoxyhaemoglobin.

Oxyhaemoglobin is oxygen-bound and bright red. Deoxyhaemoglobin has released oxygen and is dark red.

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TOPIC 10.3 — RESPIRATORY PIGMENTS

Textbook topic 10.3 · printed pages 224–225

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Q.3 Compare haemoglobin and myoglobin. How do their structures suit their functions?

Board long question · 8 marks

Feature	Haemoglobin	Myoglobin
Location	Red blood cells	Skeletal and cardiac muscle cells
Structure	Four globin chains (2α, 2β) and four haem groups	Single polypeptide chain and one haem group
Oxygen binding	Can carry up to four O_2 molecules	Can bind one O_2 molecule
Affinity	Lower affinity; releases oxygen to tissues	Higher affinity; stores oxygen in muscles
Function	Transports oxygen from lungs to tissues	Stores oxygen for muscle use

Each haem group contains an iron ion held in a porphyrin ring. Under high pO_2, iron binds oxygen. Haemoglobin's cooperative binding allows efficient loading in lungs and unloading in tissues, while myoglobin's high affinity makes it suitable for oxygen storage.

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Important Conceptual Short Questions — Topic 10.3

S1. Why does haemoglobin have four haem groups?

It has four polypeptide chains, each associated with a haem group, allowing one haemoglobin molecule to carry up to four oxygen molecules.

S2. How does myoglobin's high oxygen affinity suit its function?

It allows myoglobin to store oxygen in muscle cells and release it only when oxygen levels are very low during intense activity.

S3. Why is haemoglobin better suited for oxygen transport than myoglobin?

Its lower affinity and cooperative binding allow it to load oxygen in the lungs and unload it in tissues, while myoglobin holds oxygen too tightly for efficient transport.

TOPIC 10.4 — RESPIRATORY DISORDERS

Textbook topic 10.4 · printed pages 225–230

Q.4 Describe the major respiratory disorders, their causes, symptoms and treatment.

Board long question · 8 marks

Upper respiratory tract infections

- **Sinusitis:** Inflammation of sinus lining, usually viral. Symptoms include fever, plugged nose, pus-like discharge, facial pain and headache. Most cases resolve without antibiotics; bacterial cases may need antibiotics or nebulization.

- **Otitis media:** Inflammation of the middle ear due to fluid accumulation in the Eustachian tube. Symptoms include ear pain, fever, fluid drainage and hearing difficulty. Treatment includes pain killers and antibiotics if bacterial.

Lower respiratory tract infections

- **Pneumonia:** Alveoli become inflamed and filled with pus/fluid, limiting oxygen intake. Caused by bacteria (e.g., Streptococcus pneumoniae) or viruses. Symptoms include cough with phlegm, shortness of breath, chest pain and fever. Treated with antibiotics and prevented by vaccination.
- **Pulmonary tuberculosis:** Chronic infection by Mycobacterium tuberculosis. It damages alveoli and may form cavities. Symptoms include cough with blood, evening fever, night sweats and weight loss. Treated with multiple antibiotics for about 9 months.

Disorders of the lungs

COPD: Chronic inflammatory disease commonly caused by smoking and pollutants. Symptoms include persistent cough with mucus, shortness of breath, wheezing and fatigue. **Emphysema**, a component of COPD, involves destruction of alveolar walls, reducing surface area for gas exchange.

Important Conceptual Short Questions — Topic 10.4

S1. How does pneumonia reduce oxygen intake?

Inflamed alveoli fill with pus and fluid, reducing the surface area and diffusion of oxygen into the blood.

S2. Why is pulmonary tuberculosis treated with multiple antibiotics for a long period?

The bacteria are slow-growing and can persist in lung tissue, so prolonged combination therapy is needed to eradicate infection and prevent resistance.

S3. How does smoking contribute to COPD?

It causes chronic inflammation and damage to airways and alveoli, leading to airflow limitation and reduced gas exchange.

S4. Why does emphysema cause shortness of breath?

Destruction of alveolar walls reduces the surface area available for gas exchange, so less oxygen enters the blood.

S5. How does otitis media develop?

Fluid accumulates in the Eustachian tube and cannot drain from the middle ear, allowing bacteria or viruses to grow and cause inflammation.

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